Channel-Layer-Oriented Lightweight Spectral–Spatial Network for Hyperspectral Image Classification

Chunchao Li, Behnood Rasti, *Senior Member, IEEE*, Xuebin Tang, Puhong Duan, *Member, IEEE*, Jun Li, and Yuanxi Peng

***Abstract*— Hyperspectral image (HSI) classification is commonly influenced by convolution neural networks (CNNs). However, the large number of parameters and computational complexity associated with CNNs can limit their practical application, particularly when computing and storage resources are limited. To address this challenge, we propose a channel-layer-oriented lightweight network for HSI classification. Motivated by existing structures that typically set large channels and stack multiple layers, we give more optimal solutions strategically to further compress the model. For intralayer feature extraction, we develop a channel-oriented spectral–spatial module (COS^{2}M), which introduces a dual-single-channel (DSC) 3-D convolution that works in conjunction with depthwise convolution to fully extract spectral–spatial information. For interlayer information transmission, we propose a novel neighbor-pixel-aware activation function (NPAF), where the activation of a single pixel is determined by the learnable interaction with its neighbor range that enhances information transmission and improves the network's fitting ability through the single activation layer. By implementing these strategies, we aim to overcome the limitations of traditional CNNs and enable efficient HSI classification within resource-constrained environments. The whole network is designed to be a compact end-to-end structure. It achieves better classification performance than other deep learning methods and lightweight models, even with limited training samples. The network parameters, model complexity, and inference time also demonstrate significant superiority, as confirmed by experiments on three benchmark datasets. The source codes are available publicly at: https://github.com/AchunLee/CLOLN_TGRS**

***Index Terms*— Activation function, convolution channel, efficient model, hyperspectral image (HSI) classification, lightweight module, neural network.**

I. Introduction

THE technology of hyperspectral imaging has the capability to scan not only the spatial distribution of a target scene but also provide each pixel with hundreds of spectral features, creating 3-D cubic structured hyperspectral images (HSIs). These images are able to offer more detailed and informative data for decision making in a variety of visual tasks. HSI classification is utilized to classify each pixel into a specific class label and is commonly employed in various fields, such as geological mapping [1], [2], environmental monitoring [3], [4], change detection [5], [6], and agriculture and forestry assessment [7].

High-quality HSI classification depends on obtaining effective feature representations, typically achieved through traditional feature extraction [8] and deep learning [9]. Recently, HSI classification dramatically benefits from the exceptional multilevel feature learning ability of convolution neural networks (CNNs) [10], [11]. Roy et al. [12] combined spectral–spatial 3DCNN and spatial 2DCNN (HybridSN) to exploit the joint feature information and further learn more spatial representation. To flexibly adjust the size of the receptive field to optimize the network, the design proposed by Roy et al. [13] adopted selective 3-D convolution kernels to enhance spectral–spatial features based on improved 3-D Res-Blocks. In [14], the weighted combination of the two-way feature maps with two morphological operators was designed, which plays a vital role in the classification performance. In [15], an effective 1-D CNN was built, which mines local channel correlations layer-by-layer and comprehensively utilizes shallow and deep features to improve accurate classification. In [16], superpixel-based deformable convolution was proposed for extracting pure neighborhood spatial–spectral features. A local–global gated CNN was developed in [17] that constructs gated convolution blocks, allowing for the selection of highly discriminative information while filtering redundant information in HSI.

Moreover, there exist various deep-learning techniques for the classification of HSI, including recurrent neural networks

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Chunchao Li, Xuebin Tang, and Yuanxi Peng are with the College of Computer Science and Technology and the State Key Laboratory of High Performance Computing, National University of Defense Technology, Changsha 410073, China (e-mail: lcc@nudt.edu.cn; xbtang@nudt.edu.cn; pyx@nudt.edu.cn).

Behnood Rasti is with the Machine Learning Group, Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, 09599 Freiberg, Germany (e-mail: b.rasti@hzdr.de).

Puhong Duan is with the College of Electrical and Information Engineering and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: puhong_duan@hnu.edu.cn).

Jun Li is with the College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: lijun14@nudt.edu.cn).

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(RNNs) [18], graph neural networks (GCNs) [19], [20], and transformer [21], [22]. In [23], a novel multifeature fusion model was proposed, which incorporates multiscale full CNN and multihop GCN to extract multilevel information of HSI. Hong et al. [24] introduced groupwise spectral embeddings in transformer and effectively learned spectral local sequence information from adjacent bands of HSI to make up for the shortcomings of the classic backbone network. Zhang et al. [25] built a spatial–logical aggregation network for focusing on subtle traits extraction, where convolution operations and polymorphic transformer are used and dynamic fusing features derived from different streams. Sun et al. [26] proposed a weighted feature tokenizer for feature transformation, which can transfer CNN features and feed them into the transformer module for further feature representation and learning. Inspired by the attention mechanism, a central attention network was proposed in [27], which extracts useful information from central pixels and their similar neighborhoods by deploying scaled dot product central attention tailored for HSI. In [28], multiarea target attention was implemented to capture multiscale spectral–spatial information across areas. In addition, other specialized deep learning techniques, such as semi-supervised and unsupervised learning [29], have also been applied to HSI classification to solve problems in targeted scenarios, such as cross-scene adaption [30] and multisource fusion [31].

The methods mentioned above are primarily incremental designs or use multifeature-block fusion to improve classification accuracy, usually with high model complexity. However, as real-time demands and efficiency challenges begin to pose obstacles for hyperspectral tasks like identifying impurities in conveyor belts or monitoring forests with unmanned aerial vehicles, it is essential for hyperspectral algorithms to be efficient enough for deployment on platforms with limited storage and computing resources [32]. To address this, developers are utilizing and developing lightweight strategies [33] and efficient networks [34] for hyperspectral applications. However, some shortcomings or aspects for improvement still exist.

First, although the proposed modules are lightweight in [35] and [36], they require sophisticated and specially designed fusion mechanisms to ensure sufficient feature extraction. Second, some efficient networks use dimension reduction methods, such as principal component analysis (PCA) to preprocess images [37], [38], [39], [40]. While this can be considered as feature extraction [41] and reduce the computational load of subsequent networks, it can also limit the network's ability to learn hierarchical features from raw data automatically. Third, reducing the parameters and computation of convolution operations is the main goal of lightweight strategies like depthwise and pointwise convolution. However, some hyperparameters, such as the number of output channels in each layer and the number of layers, still have to be manually determined and significantly impact the network's overall efficiency and computational complexity. Although lightweight modules that focus on convolution have been developed, many networks still utilize large output channels or stacked multiple layers to ensure learning enough information. In [42], for example, the second part sequentially connects eight layers of lightweight units, and the output channels of the top layer reach up to 256. Similarly, in [43], the output channels of the top Ghost module also reach 72. Increasing the number of channels or layers certainly enlarges the computational complexity of the network, and few existing works concentrate on analysis and discussion on reducing them.

In order to address the issues mentioned above, we sought to create a more lightweight network that is characterized by few channels and layers. Despite its reduced complexity, the network should still perform well in terms of classification accuracy. Naturally, to ensure the learning and representation capabilities of the model, we decreased the number of channels and layers while improving the feature extraction ability of individual modules and enhancing information transmission between layers. To achieve this, our efforts and contributions in this article can be summarized as follows.

1) For overall design, we propose a lightweight network that is characterized by a structure with fewer channels and layers. This is an end-to-end compact and efficient model without complex fusion mechanisms.
2) For intralayer feature extraction, we propose a channel-oriented spectral–spatial module (COS^2M). This module is a lightweight structure that can set smaller channels, with the aim of fully extracting spectral–spatial information. We build this by introducing dual-single-channel (DSC) 3-D convolution, which works in conjunction with depthwise convolution.
3) For interlayer information transmission, instead of deploying cross-layer shortcuts or concatenation, we focus on the necessary component of each layer connection in the network, i.e., the activation function. To this end, we propose an NPAF that explores compensation mechanisms for neighbor activation values, where the activation of a single pixel is not only solely determined by its own numerical calculation but also by the learnable interaction of its neighbor range, which enhances the information transmission and improves the network's fitting ability.
4) We conducted experiments on different datasets and various compared techniques, and our proposed method achieves better classification performance, even with limited training samples. The network parameters, model complexity, and inference time also demonstrate significant superiority.

Following, Section II reviews various lightweight modules and models for traditional images and HSIs systematically. Section III presents the proposed network. Section IV details the classification results, model complexity, and parameter analysis. Section V gives the conclusion and future work.

II. Related Works of Lightweight Modules and Models

Lightweight models aim to reduce complexity while maintaining accuracy in computer vision, making them a research hotspot. Numerous classic and advanced lightweight modules and networks have been proposed. A model titled MobileNetV1 [44] introduced the concept of dividing

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traditional convolution operations into depthwise and pointwise convolution. This technique decouples the channel and spatial dimension correlations, thereby reducing model parameters. The MobileNetV2 [45] introduced the inverted residuals module, which sets fewer feature channels at both ends and more feature channels in the middle. This helps in reducing inference time. The ShuffleNetV1 proposed in [46] grouped different feature maps of the input layer and finally used channel shuffle technology to achieve feature communication. Its second version [47] proposed four criteria that are conducive for improving model computational efficiency and accuracy.

Han et al. [48] proposed a GhostNet based on the "Ghost" module, which uses a set of internal features generated first for simple linear transformation to generate several "Ghost" feature maps to match needed channels, reducing the total number of parameters required and computational complexity. MobileNetV3 [49] is obtained through neural architecture search (NAS), a technique used to design neural network structures automatically from a set of candidates called the search space. Usually, a large search space is required, thus requiring training and evaluation of many network structures, which places enormous demands on time and the number of GPUs. Some NAS-based methods, such as EfficientNet [50] and HOTNAS [51], were also designed. Recently, some new operators and lightweight designs for transformers have been proposed, such as partial convolution [52] that reduces redundant computation and memory access simultaneously, and MobileFormer [53] that combines local and global features in a bidirectional manner.

Faced with the task of HSI classification and with limited computing resources, many lightweight and efficient networks and modules targeting hyperspectral spatial–spectral features are constantly being proposed. A fast and compact 3DCNN (F3DCNN) was proposed in [38] by using a kernel function in a computationally efficient way to generate feature maps. In [54], a squeezed convolution module (SCM) was proposed to replace the conventional convolution layer, which reduced the number of parameters of the network significantly without compromising on high classification accuracy. Zhang et al. [42] implanted lightweight units based on pointwise and depthwise convolution in the 3-D network, thereby reducing the number of parameters and computational overhead and giving strategies for transfer learning. Inspired by MobileNetV3, Cui et al. [39] proposed a lightweight network based on depthwise separable convolution and added an efficient spectral attention module and improved spatial attention module. In [40], depthwise separable convolution block and fast convolution block were used to accelerate the learning of the 3-D network and then fed into 2-D CNN to extract comprehensive features. Paoletti et al. [43] migrated the architecture of the Ghost module to a CNN-based network, reducing computational costs and implementing an efficient classification method. In [55], a new lightweight and efficient search space was designed to promote spectral–spatial feature fusion for a proposed multiscale NAS. Hong et al. [56] developed a new minibatch graph convolution network to reduce the computing cost in training and explored different fusion strategies to improve classification performance. Zhang et al. [36] proposed a lightweight separable spatial–spectral self-attention module in the transformer and designed an adaptive feature fusion module to fuse the obtained spectral and spatial information.

III. Proposed Method

A. Recall Convolution Operation

Convolution operations account for the most significant computational overhead in CNNs. Denoting the feature map of a 2-D convolution layer l as $F^l \in \mathbb{R}^{W\times H\times C_{\text{in}}}$, where C_{in} indicates the number of the input channel, W and H are the width and height of the current feature map, respectively. Given a 2-D convolution kernel ($k_w \times k_h$) and setting the number of output channels to C_{out} (i.e., using C_{out} filters), the output feature map denoted by $F^{l+1} \in \mathbb{R}^{W_{\text{out}}\times H_{\text{out}}\times C_{\text{out}}}$, where the values of W_{out} and H_{out} need to be determined based on specific padding and stride settings. Specifically, in this convolution layer, the number of contained parameters can be calculated as follows:

$$P_{\text{2-D conv}} = C_{\text{in}} \times k_w \times k_h \times C_{\text{out}} \tag{1}$$

and the yielding floating point operations (FLOPs) are

$$\text{FP}_{\text{2-D conv}} = C_{\text{in}} \times k_w \times k_h \times C_{\text{out}} \times W_{\text{out}} \times H_{\text{out}}. \tag{2}$$

Due to the use of batch normalization (BN) in the following network, we do not count bias here.

In addition, consider using the computing process of 3-D convolution. The layer's input should be denoted by $F^l \in \mathbb{R}^{W\times H\times D\times C_{\text{in}}}$, which increases the depth dimension D compared with 2-D convolution. Convolution kernels shaped like ($k_w \times k_h \times k_d$) need to slide on three dimensions to obtain the output feature map $F^{l+1} \in \mathbb{R}^{W_{\text{out}}\times H_{\text{out}}\times D_{\text{out}}\times C_{\text{out}}}$. The parameters and FLOPs of the entire 3-D convolution are calculated as follows:

$$P_{\text{3-D conv}} = C_{\text{in}} \times k_w \times k_h \times k_d \times C_{\text{out}} \tag{3}$$

$$\text{FP}_{\text{3-D conv}} = C_{\text{in}} \times k_w \times k_h \times k_d \times C_{\text{out}} \times W_{\text{out}} \times H_{\text{out}} \times D_{\text{out}}. \tag{4}$$

B. Channel-Oriented Spectral–Spatial Module

We aim to utilize fewer channels cooperated with lightweight strategies for convolution layers and to complete the adequate extraction and representation of the features of HSI, consequently designing the essential COS^{2}M. The schematic of the COS^{2}M is shown in Fig. 1(a). First, we deploy a low-output channel pointwise convolution (denoting as P-conv in the following figures and table) layer to reduce the spectral dimension of HSI first. Given the hyperspectral data, $H \in \mathbb{R}^{R\times C\times B}$, R and C are the spatial size, and B represents the number of bands and the number of input channels when performing 2-D convolution. The output channel K specifies the number of dimensions to be retained and is typically much smaller than B, enabling extracting spectral information and reducing the input channels for subsequent layers. Pointwise convolution, i.e., using a 1×1 convolution kernel, can significantly reduce the computational complexity

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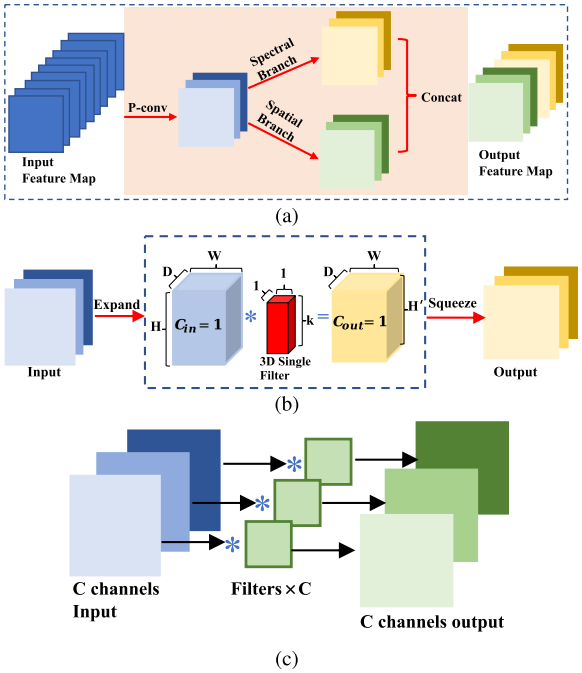

Fig. 1. (a) Schematic of the COS^{2}M. The illustrations of the operations in the spectral and spatial branches of COS^{2}M. (b) Spectral branch: DSC 3-D convolution. (c) Spatial branch: depthwise convolution.

of this convolution layer, as opposed to using larger kernel sizes such as 3×3, as demonstrated in (2). After pointwise convolution, the resulting feature map ($F \in \mathbb{R}^{R\times C\times K}$) is divided into two branches for further feature extraction.

Next, we propose a DSC 3-D convolution and expand-squeeze operation to enhance spectral feature representation, and the illustration is given in Fig. 1(b). Specifically, performing 3-D convolution, the K-presented dimension of F is considered as the depth dimension. Therefore, we should expand F to the 4-D space as $F_{\text{3-D}} \in \mathbb{R}^{R\times C\times K\times 1}$ for adding the single input channel to match the 3-D convolution pattern. Subsequently, we also set the output channel as single, together with the appropriate padding and stride size, resulting in output ($F_{\text{3-D out}} \in \mathbb{R}^{R\times C\times K\times 1}$) without changing shape. Finally, the squeeze operation is applied to obtain a feature map ($F_{\text{out}} \in \mathbb{R}^{R\times C\times K}$) of the same size as F after convolution. The design of the DSC takes full advantage of the 3-D convolution's ability to learn features automatically while minimizing computational complexity due to its few channels. In addition, by setting the 3-D convolution kernel as $1 \times 1 \times 3$, the layer becomes more specialized in extracting spectral features, further reducing computation. As per (3) and (4), the number of introduced parameters in this layer is only 3 for depth dimension, with FLOPs dramatically limited to $3 \times R \times C \times K$. Furthermore, the input and output have shape symmetry through a simple expand-squeeze operation without extra computing cost.

Another branch for spatial feature extraction involves the use of depthwise convolution [57], which is shown in Fig. 1(c). Each filter corresponds to one channel, equivalent to performing g-times convolution that configures one input and one output channel. When setting the kernel size 3×3, the parameter amount for the depthwise convolution layer is $3 \times 3 \times K$, and the FLOPs are $3 \times 3 \times K \times R \times C$, where keeping the output size unchanged with padding. It is clear that depthwise convolution significantly decreases the computational cost compared with standard convolution.

Finally, the two branches' outputs are concatenated to form a comprehensive spectral–spatial feature block denoted by $M_{\text{out}} \in \mathbb{R}^{R\times C\times 2K}$. Let us compare the convolution layer for a case where presuming has 32 input channels and 64 output channels. Using a standard convolution layer with a kernel size of 3×3 generates 18 432 parameters ($32 \times 3 \times 3 \times 64$). Then, if we use depthwise separable convolution, we get 2336 parameters ($32 \times 3 \times 3 + 32 \times 64$). The SCM in [54] has 5632 parameters ($32 \times 16 + 16 \times 3 \times 3 \times 32 + 16 \times 64$), while our COS^{2}M only has 1315 parameters ($32 \times 32 + 3 + 32 \times 3 \times 3$), making it more efficient. The network's parameters are optimized and updated by minimizing the cross-entropy loss through the back-propagation algorithm.

C. Neighbor-Pixel-Aware Activation Function

In order to introduce the information transmission of useful features through single activation functions, we propose a new design called NPAF.

Given a feature map also denoted by $F \in \mathbb{R}^{W\times H\times C_{\text{in}}}$ with width (W), height (H), and channel count (C_{in}), the map can be outputted by a convolution layer or designed module. For the conventional activation function f, activate values y' are calculated for each point x' in the map, denoted by $y' = f(x')$. This is a purely numerical calculation process without any interaction between information. On the contrary, our proposed NPAF treats a single value x' in the feature block as a pixel, and its final activation value is determined by the weighted linear combination of the activated pixels in its neighbor range. The computing process diagram of NPAF is shown in Fig. 2.

Similarly, we apply a square window that is centered around x' to define the neighbor range of x', then label all the pixels in the window as $x_1, x_2, \ldots, x_n$ in order, and n is the number of pixels. Denoting NPAF for input x' as $A(x')$, the calculation can be summarized as follows:

$$\begin{aligned} A(x') &= \sum_{i=1}^{n} w_i y_i \\ y_i &= f(x_i + b_c) \end{aligned} \tag{5}$$

where f is the based activation function, and we utilize Rectified Linear Unit (ReLU) here. In addition, we have b_c as the channel shared bias, and w_i represents the weight for each activated pixel. Unlike the self-attention mechanism, weights w do not need to be obtained by calculating the similarity relationship of the data sequence. Instead, they directly affect the activation value as the learnable parameters, automatically determining the significance in the neighbor range.

The characteristics and advantages of NPAF can be described in the following three aspects. First, from the perspective of numerical theory analysis, according to (5), NPAF can be regarded as a linear stacking of activation functions, which significantly enhances the nonlinearity and the ability of

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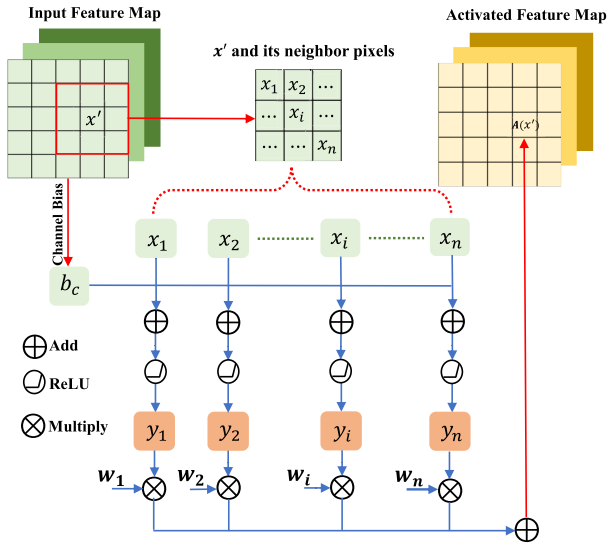

Fig. 2. Graphical overview and computing process of the NPAF. x' is the pixel to be calculated. For convenience, we use $\{x_1, x_2, \ldots, x_i, \ldots, x_n\}$ to represent all pixels in the x' neighbor window.

the network layer to fit and represent data. Another essential aspect is how the activation of a single pixel is determined. It is not only just based on its own value but also the interaction with its neighbor pixels. For instance, if ReLU is the activation function used, a pixel with a negative value will be "frozen" at 0 after activation. However, if NPAF is utilized, a pixel's value can "recover" from 0 if there are several positive values and larger weights from the neighbor pixels. This leads to a clustering correction effect that enhances the transmission of feature information. Third, compared with a traditional activation function, the NPAF introduces trainable parameters and computational complexity. However, the computational cost is relatively small when compared with convolution. In addition, the linear stacking pattern of NPAF is hardware friendly, which is great for parallel computing. Simply, the FLOPs of NPAF is

$$\mathrm{FP}_{\mathrm{npaf}} = H \times W \times C_{\mathrm{in}} \times n. \tag{6}$$

This means that the computational cost is only related to the feature map size and the number of neighbor pixels. If we set a few output channels in the previous convolution layer or module, the computational cost introduced by NPAF is also very small.

D. Channel-Layer-Oriented Lightweight Spectral–Spatial Network

Our proposal introduces a highly efficient network called a channel-layer-oriented lightweight spectral–spatial network (CLOLN). The network is designed to be "thin and shallow," consisting of only three main feature extraction layers with low output channels, i.e., two COS^{2}M and one convolution layer. A detailed network diagram is presented in Fig. 3.

The input to our model is a 3-D patch pattern that contains complete spectral information. We use two COS^{2}M modules in sequence to extract hierarchical spectral–spatial features from the 3-D block. To avoid redundancy and feature overfitting between the two COS^{2}M, we deploy conventional ReLU to introduce the numerical-based nonlinearity. In addition, we apply NPAF to improve information transmission and fitting ability in the output of the second COS^{2}M. The resulting feature block is then passed through the final 2-D convolution layer, which is also activated with NPAF. Among them, we use channel attention mechanism (CAM) modules based on squeeze and excitation block [58] to improve important channelwise feature responses. As shown in Fig. 3, CAM has a more negligible computational overhead because it involves just two fully connected layers.

IV. Experiments

A. Datasets

1) Pavia University: The Reflective Optics System Imaging Spectrometer (ROSIS-03) optical sensor captured a 103-band image of the University of Pavia campus in Italy. Pavia University (PU) dataset has a spectral coverage from 0.43 to 0.86 μm and a spatial resolution of 1.3 m per pixel. The land cover has large blocks and tiny strip areas with the image size of 610 $\times$ 340, and the corresponding ground truth contains nine classes with 42 776 labeled pixels. Fig. 4 displays the pseudocolor image, ground truth, class names, and their color illustrations.

2) WHU-Hi-HanChuan: The WHU-Hi-HanChuan (WHHC) [59] dataset was captured using a UAV-based sensor [60] over a rural–urban frontier zone in Hubei, China. It contains 274 bands from 400 to 1000 nm, with a spatial size of 1217 $\times$ 303 and a spatial resolution of about 0.109 m/pixel at an altitude of 250 m. Due to device memory constraints, only the upper half of the original image was used for classification, resulting in experimental data size of 617 $\times$ 303 $\times$ 274 and 13 classes in the ground truth, with 1 28 730 labeled pixels. Detailed graphs are shown in Fig. 5.

3) Dioni: The Dioni dataset is a part of the "HyRANK Hyperspectral Satellite Dataset I" and was released in 2018. It consists of publicly available Hyperion sensor datasets and corresponding ground truth data. The dataset has 176 spectral bands and a spatial size of 250 $\times$ 1376, with a spatial resolution of 30 m. The provided labeled samples contain 12 classes of ground objects with a scattered distribution. The false color composite image, ground truth, class names, and their color illustrations are illustrated in Fig. 6.

B. Experimental Setup

1) Comparison Methods: In order to comprehensively evaluate the effectiveness of the proposed method, several state-of-the-art deep learning techniques, including CNN-based, RNN-based, attention-based networks (comparison methods 1-4), as well as various efficient designs (comparison methods 5 and 6) and lightweight models (comparison methods 7-10) for HSI classification, are selected as comparative methods. Their brief descriptions and basic settings are listed as follows.

1) *Support Vector Machines (SVM):* The original image is classified using SVM as a baseline comparison. SVM is implemented by the libsvm toolbox and uses the Gaussian kernel, where the two hyperparameters c, ranging

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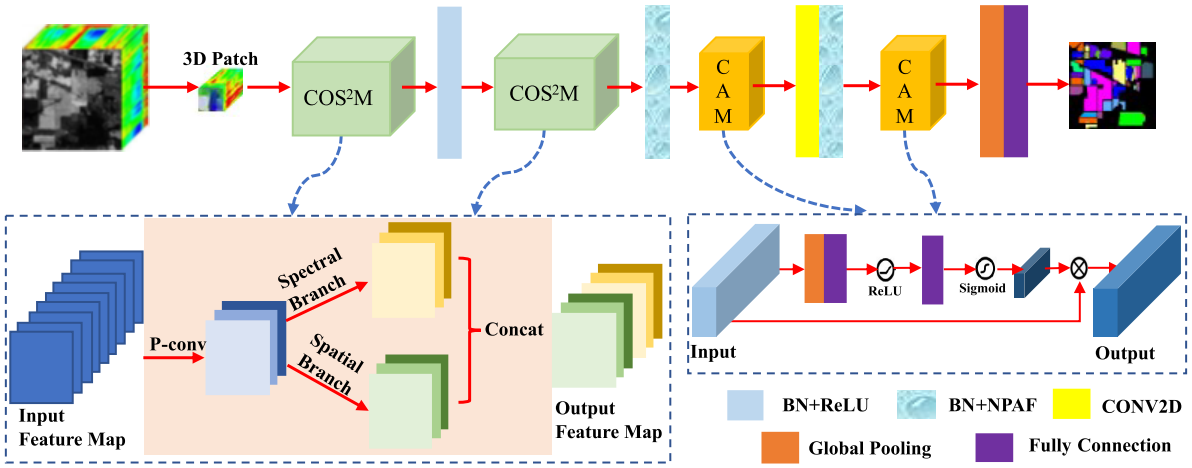

Fig. 3. Graphical overview of the proposed network. The lightweight COS^{2}M is designed for spectral–spatial feature extraction with few channels. NPAF is proposed to enhance information transmission and improve fitting ability. CAM is applied to enhance the important channelwise feature responses.

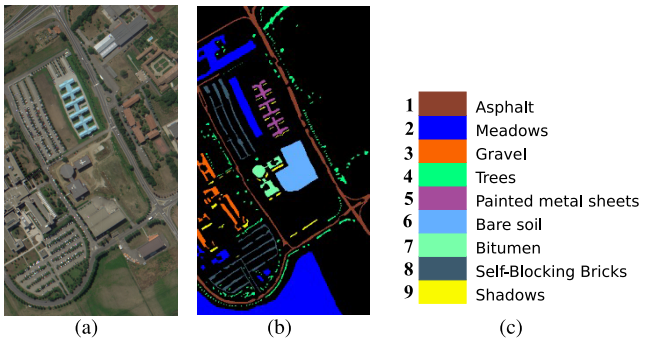

Fig. 4. PU dataset. (a) Pseudocolor image. (b) Ground truth. (c) Class names and their color illustrations.

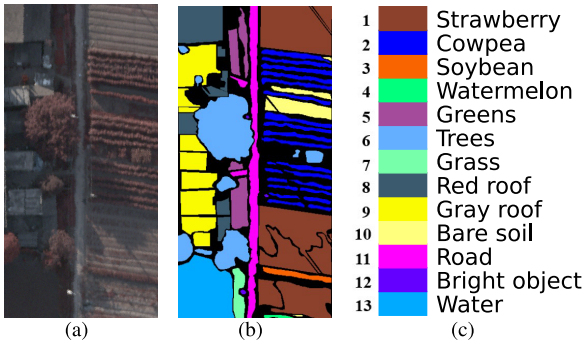

Fig. 5. WHHC dataset. (a) Pseudocolor image. (b) Ground truth. (c) Class names and their color illustrations.

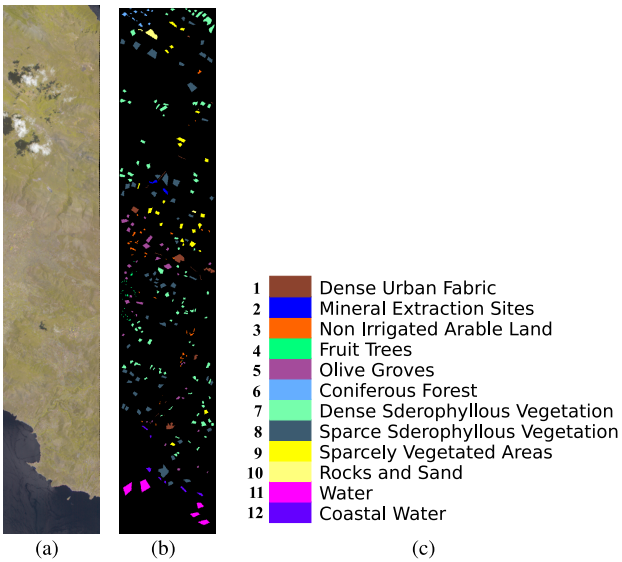

Fig. 6. Dioni dataset. (a) Pseudocolor image. (b) Ground truth. (c) Class names and their color illustrations.

from 10^{-2} to 10^4 and γ, ranging from 10^{-3} to 10^4, are determined by fivefold cross-validation.

2) *Residual Spectral-Spatial Attention Network (RSSAN) [61]:* An end-to-end trainable 2-D CNN framework, applying spectral and spatial attention to refine the useful band and "good" pixels. The method runs 200 epochs and uses the grid search to find the best learning rate (LR) from range 0.01, 0.003, 0.001, 0.0003, 0.0001, 0.00003 for each dataset. The spatial size of input 3-D patches is 9×9.
3) *SSCadRNN [18]:* A cascaded RNN model to explore the complementary spectral–spatial information of HSI by utilizing gated recursive units. The LR and training epochs are set to 0.001 and 300, respectively. The input spatial size is 27×27.
4) *HybridSN [12]:* A combination of spectral–spatial 3DCNN and spatial 2DCNN exploits the joint feature information and further learns more spatial representation. The method runs for 100 epochs with an LR of 0.001 on 3-D patches with spatial dimensions of 25×25.
5) *F3DCNN [38]:* A fast and compact 3DCNN, using a kernel function in a computationally efficient way to generate feature maps. The LR is set to 0.001, and 50 epochs are needed for training. The patch size is 11×11.
6) *A-SPN [62]:* An efficient attention-based second-order pooling network. The initial LR is determined using a swift strategy, and 9×9 is selected as the suitable patch size for the training epoch of 15.

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7) *S3EResBoF [37]:* A lightweight spectral–spatial residual bag-of-features learning model. It is trained with an LR of 0.0003 for 200 epochs with input patches kept at 15×15 in spatial dimension.
8) *Lightweight Spectral–Spatial Attention Feature Fusion Network (LMAFN) [63]:* An NAS-guided lightweight spectral–spatial attention feature fusion network. The model is trained for 1000 epochs with an LR of 0.001. The spatial patch size is set to 9×9 for optimal accuracy.
9) *Res-LS^{2}CM [64]:* A lightweight residual network that uses a lightweight spectral–spatial convolution module to build the basic residual block, with an LR of 0.0001 and a fixed input patch spatial size of 11×11, 200 epochs are chosen.
10) *Ghostnet [43]:* A lightweight CNN architecture based on a ghost module. The model is trained in about 500 epochs with an LR of 0.1. The spatial size of patches is 11×11.

All the parameters are set to the respective articles' default settings to achieve their best performance. Please refer to each reference for details.

2) Configuration and Metric: We use a small proportion of samples for training. In the PU and WHHC datasets, we randomly select 1% of available samples per class for training and validation. For the Dioni dataset, the ratio is 3%. The remaining samples constitute the test set. Average accuracy (AA), overall accuracy (OA), and Kappa coefficient (KAPPA) are used as classification performance metrics. The model parameters (Params), the number of multiply-accumulate operations (MACs), and inference time (the time of inferring the whole scene, including all the labeled and unlabeled pixels in the image) are used to evaluate the computational complexity and efficiency of the models comprehensively. Training time is also reported together. The hardware is a computer configured with Intel i7-11700, NVIDIA RTX 3060, and 32-GB RAM.

3) Parameters Setting and Analysis: For the input 3-D patches, the spectral dimension retains all bands present in the original image, and the spatial dimension determines the number of pixels in the window around the central pixel. We analyze the impact of different spatial sizes, ranging from 7 to 17 with stride 2, on the performance. Fig. 7 shows the relevant results. With an increase in the spatial size, the classification accuracies of all three datasets boost to a certain extent as more spatial information is involved in learning. However, when the size exceeds a specific value, the growth slows down, and there is a significant decrease in the PU and Dioni datasets. This is because adding excess neighboring pixels hinders the model from extracting discriminative features for the current central pixel. In addition, as the size increases, the Macs of the model steadily rise. Therefore, to maintain optimal classification performance and model efficiency, we selected a spatial patch size of 13×13 ($R = C = 13$) for the PU and Dioni datasets and applied 11×11 in WHHC.

The number of output channels for pointwise convolution in COS^{2}M is fixed at 16 ($K = 16$), and the subsequent 2-D convolution layer with 1×1 kernel size also has 16 output channels. The number of neighbor pixels in NPAF is 9 ($n = 9$). The architecture of the entire network with the output and the number of parameters of each layer are illustrated in Table I, using the PU dataset as an example. The stochastic gradient descent algorithm serves as the optimizer for the model with an LR of 0.05. The whole model runs 300 epochs, and the batch size is 32.

We also give analysis experiments of the two output channel values that must be set in the CLOLN using the PU dataset. We tested three typical values: 8, 16, and 32, and the results are shown in Table II. When the output channel of pointwise convolution in COS^{2}M and 2-D convolution layer is set to 16, the AA, OA, and KAPPA of the model reach their highest levels. A channel count of 8 may result in model parameters not being able to carry sufficient feature representations, while 32 introduces more information redundancy. The results

TABLE I
LAYER-/MODULE-BASED SUMMARY OF THE PROPOSED CLOLN ARCHITECTURE IS SHOWN IN FIG. 3 WITH PU DATASET AS AN EXAMPLE

Layer/Module	Output Shape	Parameters
Input	(103, 13, 13)	0
COS^{2}M1	(32, 13, 13)	1795
BN1&ReLU	(32, 13, 13)	64
COS^{2}M2	(32, 13, 13)	659
BN2&NPAF	(32, 13, 13)	416
CAM1	(32, 13, 13)	128
Conv2D	(16, 13, 13)	512
BN3&NPAF	(16, 13, 13)	208
CAM2	(16, 13, 13)	32
GP&FC	(9)	144
A total of 3958 trainable parameters are required		

TABLE II
COMPARISON EXPERIMENTS OF SETTING DIFFERENT OUTPUT CHANNELS OF POINTWISE CONVOLUTION IN COS^{2}M AND 2-D CONVOLUTION LAYER USING PU DATASET

Output Channel	P-conv			CONV2D		
	8	16	32	8	16	32
AA(%)	96.84	**98.50**	98.01	96.29	**98.50**	97.34
OA(%)	97.79	**99.06**	98.67	98.39	**99.06**	97.96
KAPPA(%)	97.06	**98.74**	98.24	97.86	**98.74**	97.30
Params	**2014**	3958	8806	**3510**	3958	4918
MACs(M)	**0.32**	0.64	1.42	**0.58**	0.64	0.77

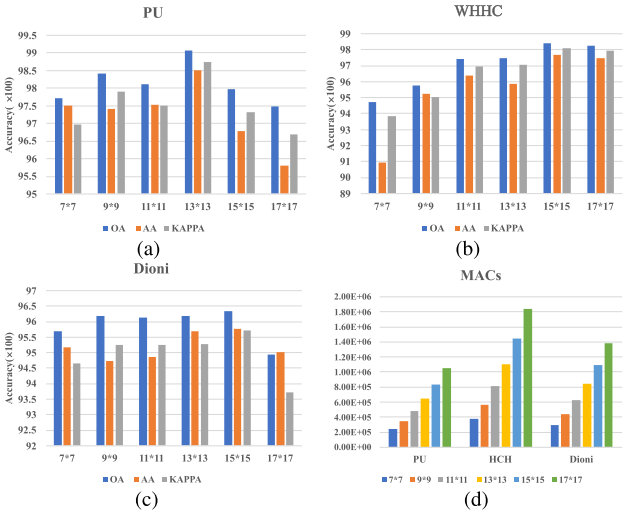

Fig. 7. Classification results for three datasets with varying patch sizes. (a) PU. (b) WHHC. (c) Dioni. (d) Model MACs are also tested.

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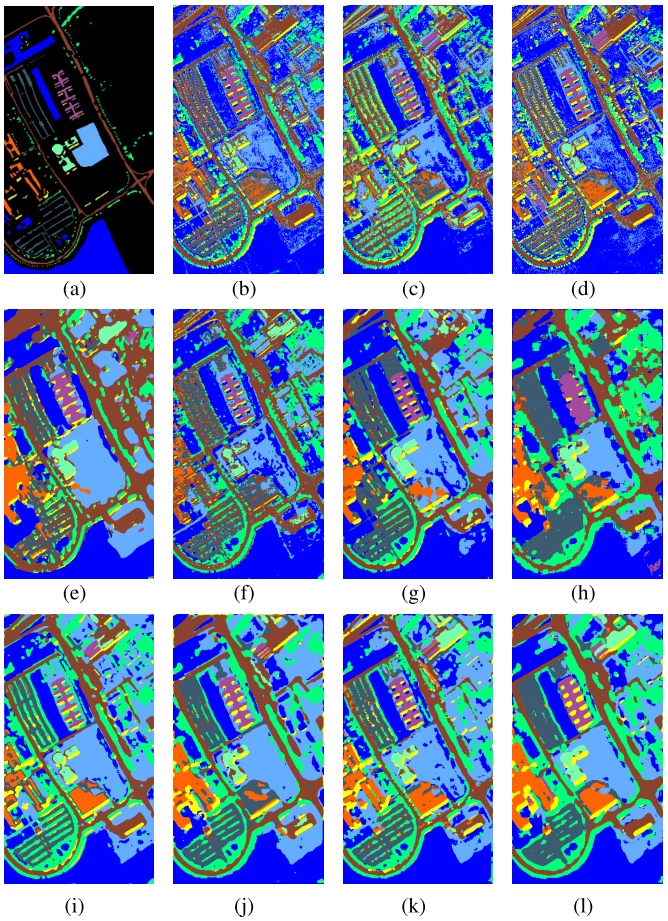

Fig. 8. Classification maps for PU dataset. (a) Ground truth. (b) SVM. (c) RSSAN. (d) SSCasRNN. (e) HybridSN. (f) F3DCNN. (g) A-SPN. (h) S3EResBoF. (i) LMAFN. (j) Res-LS^{2}CM. (k) Ghostnet. (l) CLOLN.

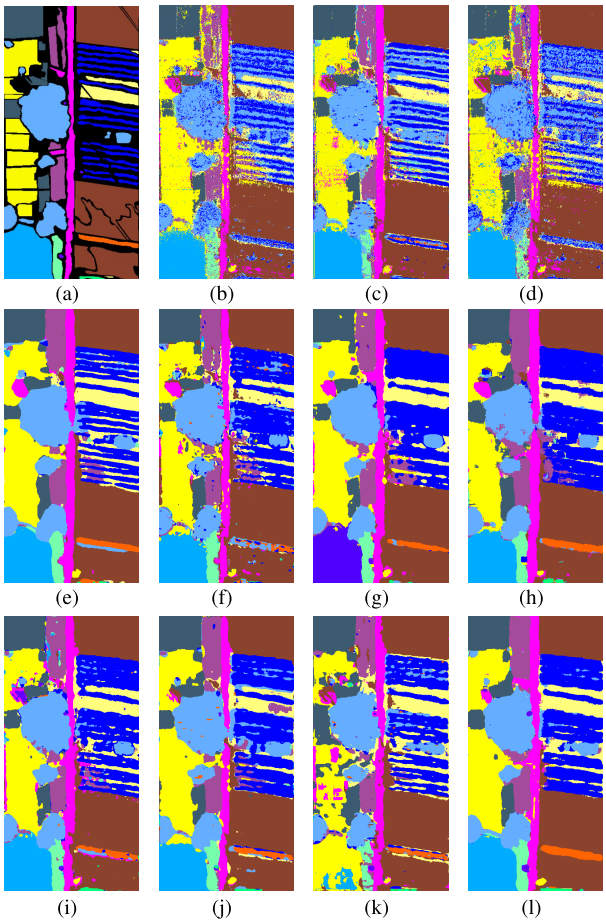

Fig. 9. Classification maps for WHHC dataset. (a) Ground truth. (b) SVM. (c) RSSAN. (d) SSCasRNN. (e) HybridSN. (f) F3DCNN. (g) A-SPN. (h) S3EResBoF. (i) LMAFN. (j) Res-LS^{2}CM. (k) Ghostnet. (l) CLOLN.

confirm that setting both channels to 16 can achieve the best balance between computational complexity and classification performance.

C. Results and Discussion

1) Classification Performance: For the PU dataset, the experimental results of each method, including OA, AA, KAPPA, and the accuracy of each class, are given in Table III, where the greatest value of each item is in bold. The specific number of train, validation, and test samples for each class is also listed in Table III. In addition, the ground-truth image and the classification maps of each method for the whole image are shown in Fig. 8. Due to the lack of feature extraction, the SVM fed with the original HSI has poor classification performance. As a result, there are many misclassifications in the classification map, which appear to be noise-like in each class. The performance of RSSAN, F3DCNN, and spectral-spatial cascaded RNN (SSCasRNN) is relatively weak because they do not undergo sufficient learning to achieve productive representation under limited training samples, and their classification maps yield quite clear errors. HybridSN and A-SPN exhibit high OA but suffer from low AA due to poor classification of certain classes, such as Class 9 in HybridSN and Class 3 in A-SPN, resulting in blocky errors in classification maps. Overall, a weak performance is observed in Classes 5 and 8 in S3EResBoF, indicating an unclear skeleton structure in the classification map compared with the ground truth. In General, LMAFN, Res-LS^{2}CM, and Ghostnet perform well in classification performance, and the noisy pixels in the classification map are reduced considerably, but LMAFN and Ghostnet struggle with Class 3, with an accuracy below 90%. Our CLOLN approach outperforms other advanced techniques with the highest AA, OA, and KAPPA scores, and all class accuracies are greater than 95%. Furthermore, the map generated by our approach is the most precise and smooth, as evidenced by the comparison with the ground-truth image.

Table IV lists the train, validation, and test numbers for each class and the results of all metrics for the WHHC dataset. It is evident that our method delivers excellent classification performance, with AA, OA, and KAPPA exceeding 96%, 97%, and 0.97, respectively-all the highest among all comparison methods. Specifically, the SVM, RSSAN, and SScasRNN models still perform poorly due to using only 1% of labeled samples for the training set, and Ghostnet's classification is not as good as its performance in the PU dataset. HybridSN, F3DCNN, A-SPN, and LMAFN all show high OA but have relatively low-AA classification results, with an accuracy below 80% for more than one class. S3EResBoF and Res-LS^{2}CM exhibit good outputs, but more than one classes have accuracy below 90%. Conversely, our CLOLN showcases excellent classification performance, with an accuracy of over 91% across 12 classes. Fig. 9 shows the ground truth and all

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TABLE III
CLASSIFICATION RESULTS, NETWORK PARAMETERS, MODEL COMPLEXITY, TRAINING TIME, AND INFERENCE TIME OF ALL STUDIED APPROACHES FOR THE PU DATASET

Class	Training set	Validation set	Test set	SVM	RSSAN	SSCasRNN	HybridSN	F3DCNN	A-SPN	S3EResBoF	LMAFN	Res-LS^{2}CM	Ghostnet	CLOLN
1	66	66	6499	86.64	88.51	87.42	90.65	94.9	91.30	97.55	98.95	96.43	**99.01**	98.99
2	186	186	18277	91.77	99.22	90.09	98.45	99.42	**99.85**	98.92	99.16	99.72	99.41	99.71
3	20	20	2059	77.74	46.68	59.74	86.14	63.09	71.24	97.43	88.02	97.40	75.62	**99.07**
4	30	30	3004	96.73	89.83	86.62	90.01	80.71	93.83	90.67	94.36	97.33	**98.44**	97.80
5	12	12	1321	99.02	86.13	98.5	99.47	**100**	**100**	88.99	99.70	99.55	99.55	99.69
6	50	50	4929	85.54	81.98	55.31	**100**	73.69	92.69	99.99	99.94	92.54	98.71	99.96
7	12	12	1306	76.83	56.8	76.46	99.92	89.6	95.19	97.54	91.64	97.31	96.51	**100**
8	36	36	3610	78.37	66.1	83.68	93	84.77	**97.62**	84.27	93.28	94.58	94.69	96.14
9	8	8	931	99.79	91.69	99.47	72.15	95.73	98.08	90.98	**99.89**	91.86	98.10	95.16
AA(%)				88.05	78.55	81.92	92.2	86.88	93.31	94.03	96.10	96.30	95.56	**98.50**
OA(%)				88.89	87.53	83.35	95.24	90.94	95.48	96.18	97.62	97.30	97.20	**99.05**
KAPPA				0.85	0.83	0.81	0.94	0.88	0.94	0.95	0.97	0.96	0.96	**0.99**
Params(K)					18.45	349.07	4844.79	994.55	96.06	245.78	156.13	10.91	43.87	**3.96**
MACs(M)					0.83	6.32	9.69	1.99	2.87	3.75	0.96	1.26	1.79	**0.64**
Training Time(s)					88.6	83.5	121.5	16.4	**3.7**	1805.1	519.2	46.4	164.1	43.4
Inference Time(s)					130.4	30.1	166.0	48.5	28.3	288.1	106.6	13.1	14.3	**11.4**

TABLE IV
CLASSIFICATION RESULTS, NETWORK PARAMETERS, MODEL COMPLEXITY, TRAINING TIME, AND INFERENCE TIME OF ALL STUDIED APPROACHES FOR THE WHHC DATASET

Class	Training set	Validation set	Test set	SVM	RSSAN	SSCasRNN	HybridSN	F3DCNN	A-SPN	S3EResBoF	LMAFN	Res-LS^{2}CM	Ghostnet	CLOLN
1	350	350	33918	91.45	98.27	92.71	98.26	99.24	99.20	98.86	98.73	97.11	96.36	**99.26**
2	141	141	13673	70.12	78.58	60.31	94.97	95.35	91.74	94.09	94.41	**97.21**	84.40	95.89
3	14	14	1297	26.97	9.16	1.28	90.03	74.41	86.2	92.15	65.74	83.17	**97.91**	97.29
4	3	3	256	14.72	22.87	3.42	67.81	39.3	34.6	**100**	90.08	79.86	99.19	99.49
5	59	59	5726	65.07	82.6	74.9	94.16	87.91	**99.13**	85.92	93.50	82.61	97.13	91.07
6	180	180	17438	76.34	89.3	71.77	98.71	90.11	93.92	**98.95**	96.26	95.18	96.64	97.02
7	18	18	1762	74.96	85.42	75.86	97.87	97.63	**99.22**	85.54	89.48	98.23	89.78	86.71
8	105	105	10201	93.68	95.64	93.46	95.21	99.12	96.7	97.98	98.41	98.26	**99.20**	98.94
9	169	169	16404	85.65	92.31	91.34	97.74	94.57	**99.83**	95.95	98.46	91.15	85.60	97.52
10	43	43	4208	72.28	80.21	68.84	94.69	91.36	**95.34**	92.36	93.71	93.52	59.15	91.86
11	89	89	8628	89.97	82.42	75.41	93.35	87.52	95.08	96.50	81.83	**99.45**	68.86	97.80
12	2	2	133	**100**	8.09	4.35	9.63	2.74	72.26	**100**	69.34	**100**	**100**	**100**
13	116	116	11219	98.06	96.35	96.47	**100**	96.88	99.97	99.67	98.94	99.41	98.45	99.85
AA(%)				73.79	70.86	62.32	87.11	81.24	89.48	95.23	89.91	93.47	90.20	**96.36**
OA(%)				84.27	89.91	82.17	96.92	94.57	96.87	96.61	95.77	95.39	89.56	**97.40**
KAPPA				0.82	0.88	0.80	0.96	0.94	0.96	0.96	0.95	0.94	0.88	**0.97**
Params(K)					49.71	350.09	4845.31	995.07	977.26	216.91	167.59	13.90	69.36	**6.76**
MACs(M)					3.75	15.12	9.69	1.99	2.93	3.75	1.81	1.59	3.79	**1.11**
Training Time(s)					322.1	494.4	331.4	50.8	**29.0**	2050.9	252.1	177.8	590.4	224.1
Inference Time(s)					212.3	27.9	136.5	53.0	73.1	269.0	89.5	25.4	28.1	**22.6**

classification maps. It is apparent that the classification maps of other methods contain significant error points. Our method produces the most accurate classification results with minimal misclassification of pixels.

The experimental results for the Dioni dataset are shown in Table V and Fig. 10. Our method continues to maintain the highest classification accuracy and performs better than other methods, particularly for the AA. Likewise, CLOLN achieves an accuracy of over 90% across all 13 classes, while in the other compared methods, more than one class has an accuracy lower than 90% or worse. Precise classification of each ground object is crucial for practical applications, and CLOLN has significant potential and advantages for achieving it.

What is more, starting from the initial training samples and gradually increasing them, the AA results of each model under different training ratios are demonstrated in Fig. 11. It is evident that with an increase in the number of training samples, the accuracy of all models progressively enhances and approaches a stable zone, which is a common scenario in remote sensing. Our algorithm basically maintains the best model accuracy, especially when the training ratio is low, which shows significant superiority.

2) Model Complexity: Similarly, Tables III–V display the model complexity, training time, and inference time, revealing that the proposed CLOLN achieves the most lightweight design in terms of Params and MACs. In the PU dataset, CLOLN requires only 3958 parameters for training, which is significantly lower than the parameters required by other comparison methods. The difference is so significant that even other lightweight methods require several times or even several orders of magnitude more parameters than our model. This means that our method greatly reduces memory usage and initialization costs, which is especially useful for devices with limited storage resources. In addition, the entire model requires only 0.64 MACs and delivers a substantial benefit in terms of inference speed, making it superior to other lightweight models

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TABLE V
CLASSIFICATION RESULTS, NETWORK PARAMETERS, MODEL COMPLEXITY, TRAINING TIME, AND INFERENCE TIME OF ALL STUDIED APPROACHES FOR THE DIONI DATASET

Class	Training set	Validation set	Test set	SVM	RSSAN	SSCasRNN	HybridSN	F3DCNN	A-SPN	S3EResBoF	LMAFN	Res-LS^{2}CM	Ghostnet	CLOLN
1	37	37	1188	71.03	63.95	59.10	86.19	79.66	90.62	**96.09**	87.66	86.85	88.62	90.73
2	6	6	192	68.29	28.42	39.90	58.08	28.28	99.50	98.77	96.45	99.48	**100**	97.21
3	18	18	578	71.70	31.34	66.28	80.87	74.66	85.23	83.00	**95.97**	93.78	90.37	91.54
4	4	4	142	39.44	4.2	28.77	21.92	32.19	9.52	79.91	71.72	**98.08**	66.67	90.11
5	53	53	1662	82.76	87.77	78.02	**98.61**	87.41	96.74	88.24	94.40	78.03	95.51	93.10
6	11	11	339	95.78	84.27	84.46	**100**	98.00	**100**	95.76	97.14	**100**	99.09	97.10
7	151	151	4733	90.31	91.32	89.48	93.28	95.54	95.52	96.15	92.38	**97.90**	94.25	94.46
8	191	191	5992	85.26	95.25	90.01	98.48	96.77	96.80	94.64	98.24	92.62	95.07	**98.86**
9	52	52	1650	78.69	84.98	65.92	91.01	85.54	**99.65**	94.69	94.53	88.91	96.03	96.49
10	15	15	462	89.70	76.46	94.50	94.97	90.36	89.73	99.20	98.74	89.94	98.46	**100**
11	48	48	1516	95.27	99.01	99.94	**100**	99.81	**100**	**100**	**100**	99.86	97.44	98.57
12	12	12	374	95.16	70.9	86.82	**100**	99.48	**100**	98.57	97.41	98.96	99.70	**100**
AA(%)				80.28	68.15	73.6	85.28	80.64	88.61	93.75	93.72	93.70	93.43	**95.68**
OA(%)				85.46	86.59	83.84	94.32	91.87	95.55	94.58	95.27	92.33	94.71	**96.18**
KAPPA				0.82	0.83	0.80	0.93	0.90	0.94	0.93	0.94	0.91	0.93	**0.95**
Params(K)					31.01	349.84	4845.18	994.94	372.59	216.84	161.19	12.27	55.03	**4.7**
MACs(M)					1.27	10.21	9.69	1.99	1.12	3.75	1.51	1.40	2.65	**0.77**
Training Time(s)					120.0	89.6	157.0	21.4	**8.0**	979.5	238.1	68.1	217.8	76.2
Inference Time(s)					257.2	51.3	272.6	81.9	83.8	463.8	212.9	29.0	29.7	**25.6**

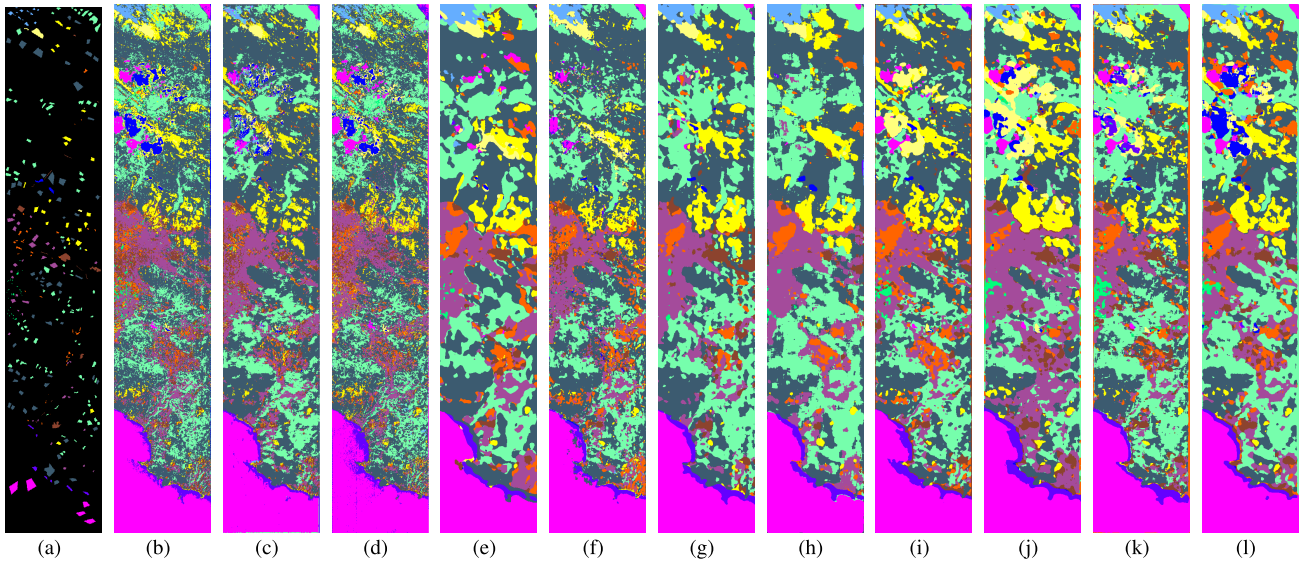

Fig. 10. Classification maps for Dioni dataset. (a) Ground truth. (b) SVM. (c) RSSAN. (d) SSCasRNN. (e) HybridSN. (f) F3DCNN. (g) A-SPN. (h) S3EResBoF. (i) LMAFN. (j) Res-LS^{2}CM. (k) Ghostnet. (l) CLOLN.

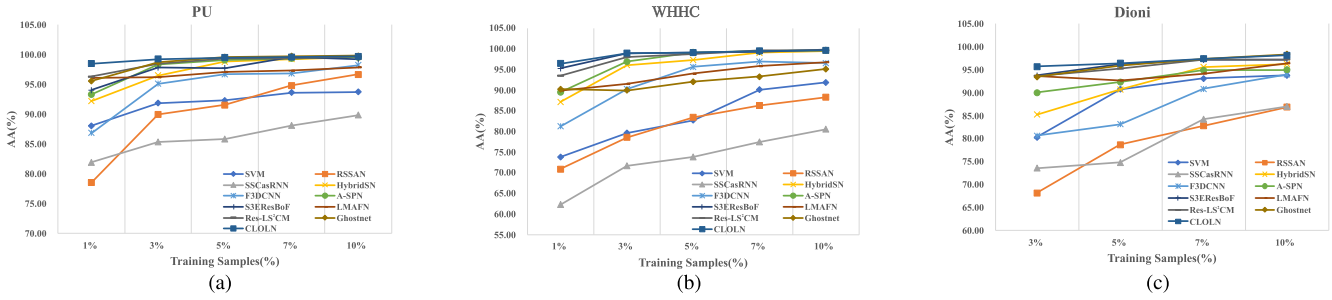

Fig. 11. Classification results of each model under different training ratios on the three datasets. (a) PU. (b) WHHC. (c) Dioni.

and efficient designs. This highlights the practicability of the model in scenarios with limited computing resources.

Our model retains all the bands of input into the network, which results in increased parameters and MACs in the WHHC dataset compared with those in the PU dataset. Despite this, CLOLN also demonstrates significant advantages in terms of model complexity and inference time compared with all other methods. For the Dioni dataset, our model maintains efficient inference speed for all pixels of this larger sized image, showcasing great hardware adaptability; model complexity still

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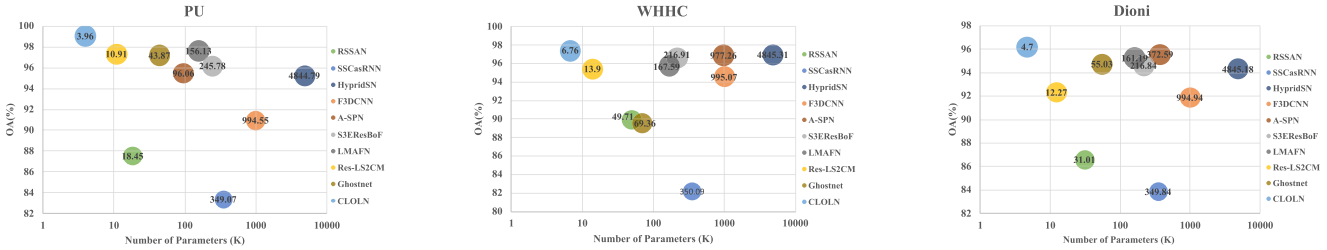

(a) (b) (c)

Fig. 12. Scatter plots of (a) PU, (b) WHHC, and (c) Dioni, where OA shown as the vertical axis and the number of parameters as the horizontal axis. The horizontal axis uses a logarithmic scale with a base of 10 for clear display.

outperforms other algorithms. Although RSSAN, F3DCNN, A-SPN, LMAFN, Res-LS^{2}CM, and Ghostnet have the same or even smaller input patch sizes as us, our CLOLN still has the smallest MACs and the fastest inference speed.

It is worth mentioning the training time of all the models, as it is directly related to the suitable training epoch required to achieve a good model of each algorithm. For instance, A-SPN and F3DCNN are set to 15 and 50 epochs, respectively, resulting in a relatively short training time. Although the training time of the model is not a direct indicator of its applied efficiency, our training time also demonstrates relatively better efficiency compared with some methods that set lower training epochs than ours (such as RSSAN, HybridSN, and S3EResBoF, they set 200 or 100 epochs, and we have 300 epochs).

In addition, Fig. 12 shows a visual comparison of the OA and model parameters of all deep learning algorithms. The scatter plots show OA as the vertical axis and the number of parameters as the horizontal axis. The horizontal axis uses a logarithmic scale with a base of 10 for clear display, and the specific value of the parameter quantity is shown in numbers at each point. Our method obviously outperforms other algorithms in both the PU and Dioni datasets in terms of classification performance and lightweight model aspects. It achieves the best classification results with smaller training parameters. In the WHHC dataset, although HypridSN, S3EResBoF, and A-SPN approach our performance in OA, their model parameters are significantly larger than ours by more than two orders of magnitude, and they also have relatively low AA.

D. Ablation Experiment and Analysis

We select the PU dataset for ablation experiments and analysis. We replace COS^{2}M in our network with other layers and lightweight modules, including a conventional 2-D convolution layer with kernel 3 (Conv2D), implementing pointwise and depthwise convolution sequentially (PDC), SCM, and LS^{2}CM, and ensure the same output channels for each layer and module. The experimental results are presented in Table VI. Our proposed module significantly outperforms the conventional convolution layer in terms of efficiency, reducing the number of model parameters and computation to about one-tenth while improving classification performance. This highlights its ability to extract features efficiently, especially in lightweight design. PDC can be considered removing spectral branch in COS^{2}M. After comparing the results between the PDC and the proposed module, it can be observed that our implementation of DSC significantly improves classification accuracy with fewer parameters and computations. Our method is also superior to SCM in terms of accuracy and efficiency. We achieved a notable improvement in classification performance, particularly in terms of AA, with only a six-parameter increase compared with LS^{2}CM. Though there is an additional computation of 0.1 M, it does not significantly affect inference time, making it hardware friendly.

TABLE VI

ABLATION EXPERIMENTS OF REPLACING COS^{2}M WITH NORMAL CONVOLUTION LAYER AND OTHER LIGHTWEIGHT MODULES FOR THE PU DATASET

Modules	AA(%)	OA(%)	KAPPA(%)	Params	MACs(M)	Inference Times(s)
Conv2D	97.82	98.52	98.45	40384	6.71	15.4
PDC	97.28	98.42	97.90	6400	1.04	12.5
SCM	97.31	98.15	97.55	5144	0.83	13.1
LS^{2}CM	97.46	98.49	98.01	**3952**	**0.63**	12.5
Proposed	**98.50**	**99.05**	**98.85**	3958	0.64	**11.4**

TABLE VII

ABLATION EXPERIMENTS OF IMPLEMENTING COS^{2}M WITH SETTING DIFFERENT KERNEL SIZE (115 AND 117) AND OUTPUT CHANNEL (2 AND 3) IN DSC FOR THE PU DATASET

Modules	AA(%)	OA(%)	KAPPA(%)	Params	MACs(M)	Inference Times(s)
COS^{2}M(2)	96.86	98.52	98.04	4876	0.79	12.4
COS^{2}M(3)	97.49	98.54	98.07	5858	0.94	11.7
COS^{2}M(115)	96.94	98.19	97.60	3962	0.65	12.0
COS^{2}M(117)	97.14	98.30	97.74	3966	0.66	11.8
Proposed	**98.50**	**99.05**	**98.85**	3958	0.64	**11.4**

Furthermore, the ablation experiments of varying the 3-D convolution kernels and output channels in DSC are conducted. The outcomes are listed in Table VII, demonstrating that multiple output channels result in lower classification performance and introduce more complexity, thus highlighting the superiority of the DSC structure. In addition, the results find our setting the convolution kernel size of $1 \times 1 \times 3$ is the optimal solution.

What is more, we provide ablation experiments and analysis of NPAF. We tried several popular activation functions, such as ReLU, PReLU (also with learnable parameters), LeakyReLU (with a negative slope of 0.1), and Hardswish to replace NPAF in our network. We also considered the parameter n settings of 25 (5×5) and 49 (7×7) in NPAF, and all the experimental results are presented in Table VIII. From the results, we can see that our NPAF significantly improves classification performance compared with other activation functions, which fully demonstrates its effectiveness and superiority. Although it introduces more parameters and computational amounts, the

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TABLE VIII
ABLATION EXPERIMENTS OF IMPLEMENTING NPAF WITH SETTING DIFFERENT PARAMETERS AND OTHER ACTIVATION FUNCTIONS FOR THE PU DATASET

Modules	AA(%)	OA(%)	KAPPA(%)	Params	MACs(M)	Inference Times(s)
ReLU	96.91	98.42	98.17	**3430**	**0.55**	12.6
PReLU	94.39	96.94	95.95	3433	**0.55**	12.2
LeakyReLU	95.88	97.18	96.28	**3430**	0.57	12.4
Hardswish	94.30	96.89	95.89	**3430**	**0.55**	12.8
NPAF (n=9)	**98.50**	**99.05**	**98.85**	3958	0.64	**11.4**
NPAF (n=25)	97.88	98.68	98.26	4726	0.77	12.6
NPAF (n=49)	96.50	98.12	97.51	5878	0.97	12.0

increase is relatively small. Moreover, from the perspective of inference time, the slight increase in computation can be optimized in current computing operators. In addition, we found that choosing 9 (3×3) for n is the optimal solution, indicating that a larger neighbor range means redundancy of information and an increase in computational complexity.

V. CONCLUSION

To address the challenges of limited computing and storage resources in hyperspectral application scenarios, we presented CLOLN, a specialized design that prioritizes setting fewer channels and layers while implementing lightweight strategies. What is more, we also developed a new module called COS^{2}M, which utilizes a DSC 3-D convolution in combination with depthwise convolution to extract spectral–spatial information more effectively. In addition, we proposed a unique function called NPAF which determines the activation of a single pixel by using information from its neighbor range, resulting in improved information transmission and better fitting of the network through a single activation layer. This approach offers several advantages, including improved classification performance and model efficiency when compared with other advanced deep learning algorithms and lightweight networks. Furthermore, we demonstrate the effectiveness and superiority of the proposed COS^{2}M and NPAF through comparative experiments.

We also identify some areas that require improvement. For instance, our inference time did not demonstrate the same sufficient advantage as MACs in the comparison. This could be due to the frequent data access and memory conversion, which may hinder the computing pipeline from reaching its roofline significantly as the sample size increases. Therefore, we plan to optimize our approach in conjunction with hardware architecture, aiming to reduce the inference time of the model even further and produce more hardware friendly networks in our future work.

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Chunchao Li received the B.Sc. degree from the Huazhong University of Science and Technology, Wuhan, China, in 2018, and the M.S. degree from the National University of Defense Technology, Changsha, China, in 2020, where he is currently pursuing the Ph.D. degree with the State Key Laboratory of High Performance Computing.

His research interests include remote sensing image processing, parallel computing, and intelligent perception.

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Behnood Rasti (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electronics and electrical engineering from the Electrical Engineering Department, University of Guilan, Rasht, Iran, in 2006 and 2009, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Iceland, Reykjavik, Iceland, in 2014.

In 2015 and 2016, he worked as a Post-Doctoral Researcher with the Electrical and Computer Engineering Department, University of Iceland. From 2016 to 2019, he was a Lecturer with the Center of Engineering Technology and Applied Sciences, Department of Electrical and Computer Engineering, University of Iceland. He is currently a Principal Research Associate with Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany. His research interests include signal and image processing, machine/deep learning, remote sensing, and artificial intelligence.

Dr. Rasti was a Humboldt Research Fellow in 2020 and 2021. He won the Doctoral Grant of the University of Iceland Research Fund "The Eimskip University Fund" and the "Alexander von Humboldt Research Fellowship Grant" in 2013 and 2019, respectively. He was a Valedictorian as an M.Sc. Student in 2009. He serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL).

Xuebin Tang received the B.Sc. degree in electronic and information engineering and the M.S. degree in computer science and technology from Guangxi University, Nanning, China, in 2017 and 2020, respectively. He is currently pursuing the Ph.D. degree in electronic science and technology with the State Key Laboratory of High Performance Computing, National University of Defense Technology, Changsha, China.

His research interests include hyperspectral image processing, intelligent perception, deep domain adaptation, and deep neural network acceleration.

Puhong Duan (Member, IEEE) received the B.S. degree from Suzhou University, Suzhou, China, in 2014, the M.S. degree from the Hefei University of Technology, Hefei, China, in 2017, and the Ph.D. degree from Hunan University, Changsha, China, in 2021.

From October 2019 to December 2019, he was a Visiting Ph.D. Student with the Faculty of Electrical Engineering and Computer Science, Technische Universitt Berlin, Berlin, Germany. From January 2020 to October 2020, he was a Visiting Ph.D. Student with Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany. He is currently an Associate Research Fellow and a Post-Doctoral Fellow with the College of Electrical and Information Engineering, Hunan University. His research interests include image restoration, multimodal image fusion, and classification.

Dr. Duan was the Finalist for the Best Student Paper Award at the International Geoscience and Remote Sensing Symposium (IGARSS) 2020 and selected as the Best Reviewer for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2021.

Jun Li received the B.S. degree in microelectronics from Sichuan University (SCU), Chengdu, China, in 2014, and the M.S. degree in microelectronics and the Ph.D. degree in electronic science and technology from the National University of Defense Technology (NUDT), Changsha, China, in 2017 and 2020, respectively.

He was a Visiting Student with the Department of Electronics and Computer Engineering, University of Toronto, Toronto, ON, Canada, from 2018 to 2020. He is working at the Comprehensive Situational Awareness Group, College of Electronic Science and Technology, NUDT. His research interests include intelligent perception, hyperspectral image processing, multisource data fusion, and embedded systems and applications.

Yuanxi Peng was born in 1966. He received the B.S. degree in computer science from Sichuan University, Chengdu, China, in 1988, and the M.S. and Ph.D. degrees in computer science from the National University of Defense and Technology (NUDT), Changsha, China, in 1998 and 2001, respectively.

He was a Visiting Professor with the Department of Electronic and Computer Engineering, University of Toronto, Toronto, ON, Canada, from 2010 to 2011. He has been a Professor with the Computer School, NUDT, since 2011. His research interests include hyperspectral image processing, high-performance computing, multi- and many-core architectures, on-chip networks, cache coherence protocols, and architectural support for parallel programming.

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